DEWDRIVE · STEM CHECK

DewDrive STEM check

Design Molecule Lab · designmolecule.com

DOCUMENT CONTROL

Document ID	DWD-STM-001	Version	0.2
Status	Draft	Date	2026-10-25
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/dewdrive		
Source	docs/01b-stem-check.md @ 4f86e9e		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Amish's decisions of 2026-10-25: S16 at the 65 % RH cut-out (item 3), S18 at the 25 m/s design gust (item 4)
0.1	2026-10-25	Amish Chadha	First STEM check (kit 1.8 upgrade of a TRL 3 repo); five detail failures fixed under Amish's standing instruction of 2026-10-23 (DWD-DEC-001 D16 to D20), two of them found by the independent review; independent review recorded

DewDrive STEM check

Verdict: the working principle (a desiccant bed that adsorbs from night air and is regenerated by the sun in a sealed box with a shaded condenser) sits within known physics and chemistry, but the design as drawn had five details that did not do what the calculation note assumed, and fixing them leaves the yield below its target. The worst was optical: the sun fell on white sorbent beads, not on the black surface DWD-CAL-001 had taken, and the design would have made 0.01 L per day (S5). The independent review then found the second largest: the tray floor passed about 25 % of the bed area to the vapour, not the 60 % taken (S9). The others were bare aluminium and stainless steel in hot $CaCl_2$ brine (S20), bare aluminium fins treated as a high-emissivity radiator (S12) and ground anchors with a 6 % margin (S18). All five are fixed under Amish's standing instruction of 2026-10-23 (DWD-DEC-001, D16 to D20; DWD-DDR-005). With the fixes the design collects **0.44 L per day at the design point, below the 0.5 L target of PR-001** (DWD-CAL-001 v0.8, C3; 0.37 to 0.41 L under the sensitivities), and the ways to recover it are for Amish (DWD-DEC-001, open item 2). Of the 27 claims on option A, **18 are within known science and 9 are plausible**: S16 is within since Amish lowered the fan cut-out to 65 % RH on 2026-10-25 (DWD-DEC-001 item 3); S22 and S26 are closed by calculation, and S8, S11, S18, S21, S23, S24 and S27 need a test, a log or a supplier's rating (TRL 4). `stem_check` is therefore **open**. Options B and C sit within known science, but option B makes no water at the design point and option C makes 0.2 to 0.5 L per day (rows B1 to C3). Method: STANDARDS section 20. Calculations: `docs/04-calcs/stem_check.py` , which reads the tagged results of `docs/04-calcs/sizing.py` (DWD-CAL-001 v0.8) and checks them against independent first-order estimates.

Claims checked

Table 1. Claims, governing science and verdicts (generated from `docs/04-calcs/stem_results.csv`). Values are estimates.

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S1	Night air at 20 C and 40 % RH carries about 6.92 g/m3 of water vapour [A1]	Physics	Thermodynamics	Ideal gas law on the saturation pressure (Magnus form); independent of the CAL's IAPWS fit	claimed 6.92 g/m3, estimate 6.9 g/m3 (ratio 1.00)	Within known science	DWD-CAL-001 A1
S2	Overnight uptake of the 4 kg bed (0.54 kg [B5]) stays below its equilibrium capacity at 40 % RH (1.0 kg CaCl2 as a 40.1 wt % solution plus 0.08 kg/kg per unit activity on 3.0 kg of gel)	Chemistry	Physical chemistry (equilibria)	CaCl2 solution water activity (Conde 2004); salt in the pores held as solution above deliquescence	0.466295 kg <= 1.589 kg	Within known science	DWD-CAL-001 A4, B5
S3	The fan's night air supplies more vapour than the bed takes up (2.77 kg carried [B4] against 0.54 kg adsorbed [B5])	Physics	Fluid mechanics	Mass balance on the air stream	in 2.759 kg, out 0.4663 kg	Within known science	DWD-CAL-001 B4, B5
S4	Drawing 40 m3/h down through the 9.3 mm beds costs about 0.16 Pa [B3], well within a 2 W, 120 mm fan (about 3.3 Pa for the whole box [F3])	Physics	Fluid mechanics	Ergun equation, 3.5 mm beads, bed voidage 0.40	claimed 0.162603 Pa, estimate 0.174 Pa (ratio 0.94)	Within known science	DWD-CAL-001 B3, F3
S5	The sun is absorbed on a black surface over the beds with transmittance-absorptance about 0.70 [C11]. Was 0.74 on the bare beads, which are white (absorptance about 0.3): the v0.7 design would have made 0.03 L	Physics	Optics and radiation	Transmittance of 10 mm twin-wall polycarbonate (about 0.78) times absorptance; a 22.7 % open black plate with light through the holes scattered back onto its underside; cannot exceed the plain	0.696735 <= 0.741	Within known science	DWD-DEC-001 D16; DWD-CAL-001 C11

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	per day [C11]; fixed with black perforated absorber plates on the beds			black plate (0.78 x 0.95)			
S6	Day energy balance closes: absorbed 3.46 kWh [C7] covers desorption, loss through the glazing and heat to the condenser	Physics	Heat transfer	First law over the day; the remainder is sensible heat left in the bed at dusk	in 3.464 kWh, out 3.384 kWh	Within known science	DWD-CAL-001 C7
S7	Desorbing the night's water takes 2.58 MJ/kg, at least the latent heat of water	Chemistry	Physical chemistry (thermochemistry)	Heat of desorption of a salt solution is at least the latent heat (2.26 MJ/kg at 100 C)	2.57695 MJ/kg >= 2.26 MJ/kg	Within known science	DWD-CAL-001 C7
S8	Solar-to-water efficiency of 5.4 % [C7] for a single-stage sorption collector	Physics	Thermodynamics	Published outdoor results: about 9 % for a dual-stage zeolite device (LaPotin et al., Joule, 2021); a single stage recovers no latent heat, so it should sit below that; the independent review recalls single-stage devices at a few per cent (unverified)	5.4 % below about 9 %, but high for a single stage; rests on the estimated absorber figures	Plausible, to confirm at TRL 3	Open (TRL 4)
S9	Vapour crosses the gap and the drip screen to the condenser at about 96 g/h at its peak (C2); the tray floor lets 58 % of the bed area through. Was 60 % for a 50 % perforated sheet under a fine mesh,	Physics	Heat transfer	Stefan diffusion through still air over the open share of the tray floor; hot over cold and moist (light) over dry is stably stratified; estimate taken at the noon state of the v0.8 day table	claimed 96.2075 g/h, estimate 106 g/h (ratio 0.91)	Within known science	DWD-DEC-001 D19; DWD-CAL-001 C2

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	about 25 % in fact (independent review); fixed with an open floor of bars and a coarser mesh						
S10	Condensation happens: at noon saturation at the condenser plate (value) is below the bed's vapour pressure (limit)	Physics	Thermodynamics	Vapour pressure of the bed (water activity x saturation at bed temperature) against saturation at the plate	10.5597 kPa <= 39.56 kPa	Within known science	DWD-CAL-001 C2
S11	Fins shed heat by natural convection at about 4.7 W/(m2 K) [C8], holding the condenser within 15 K of the air	Physics	Heat transfer	Laminar isolated vertical plate, Nu = 0.59 Ra^0.25, gives 5.0 W/(m2 K); but the fins hang under a hot, tilted, downward-facing floor that traps warm air (independent review): 30 % less convection gives about 14 K, and also half the radiation about 16 K	PR-005 at risk; condenser temperature to be logged at TRL 4	Plausible, to confirm at TRL 3	Open (TRL 4)
S12	The finned underside also radiates to the shaded ground as an envelope of emissivity 0.85 (C8). Bare aluminium fins (emissivity about 0.1) would not; fixed by painting the fins and plate underside with a matt high-emissivity paint	Physics	Optics and radiation	Radiation from a grey envelope, h_r = 4 eps sigma T^3; painted metal about 0.9	0.85 emissivity <= 0.9 emissivity	Within known science	DWD-DEC-001 D18
S13	On a peak day (1000 W/m2, 40 C air) a dry bed	Chemistry	Materials and polymer chemistry	Polycarbonate service rating applies to the	106.419 C <= 130 C	Within known science	DWD-CAL-001 D6

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	stagnates near 133 C and the glazing inner skin near 106 C [D6], below the 130 C rating of the specified glazing (was checked as the bed at 785 W/ m2; independent review)			sheet; inner skin from the film model of DWD-CAL-001 D2; DWD-DEC-001 decision of 2026-10-02			
S14	Surfaces touched in normal use stay at or below 60 C: glazing outer skin 55 C, fins 47 C [D2, D3]	Biology	Human physiology and biomechanics	PR-013 touch limit of 60 C; contact burn thresholds depend on material and contact time (EN ISO 13732-1, values to confirm)	54.6082 C <= 60 C	Within known science	DWD-CAL-001 D2, D3
S15	After one 90 % RH night the salt solution fills 47 % of the gel's pores [E3], so no brine drains out	Chemistry	Physical chemistry (equilibria)	Solution volume (Conde activity, solution density) against 3.0 L of pore volume	0.471179 of pore volume <= 1 of pore volume	Within known science	DWD-CAL-001 E3
S16	The fan's 65 % RH cut-out (restart below 60 %) sits below the 71 % RH at which the pores fill at equilibrium [E2]	Engineering and technology	Control	Logger rule against the pore-filling humidity; lowered from 70 % RH, 1 point below pore filling and inside a low-cost sensor's tolerance, by Amish on 2026-10-25 (DWD-DEC-001 item 3)	0.65 RH <= 0.7101 RH	Within known science	DWD-CAL-001 E3 to E5; DWD-DEC-001 item 3
S17	The 10 W panel covers the fan and logger: 23.6 Wh per day of load [F1]	Engineering and technology	Power and link budgets	Daily energy budget	in 35 Wh, out 23.6 Wh	Within known science	DWD-CAL-001 F1, F2
S18	Two ground anchors each rated 250 N in uplift hold	Physics	Mechanics and statics	Dynamic pressure, force coefficient 1.2 on	234.381 N <= 250 N	Plausible, to	

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	the unit in a 25 m/s gust; each must carry 234 N [H4] (96 N at the earlier 20 m/s [H5]). Was 100 N, a 6 % margin at 20 m/s (independent review)			1.1 m2, overturning about the downwind legs with a factor of 1.5; anchor rating in the site soil to confirm when bought; design wind speed a 25 m/s gust until the site value is known (decided by Amish 2026-10-25, DWD-DEC-001 item 4)		confirm at TRL 3	
S19	The 20 x 5 mm deck bars carry the trays, sorbent, water, absorbers and screens (about 52 N on the centre bar over 1,016 mm)	Physics	Strength of materials	Simply supported bar, sigma = M / (I / c); 6063-T5 aluminium yield about 110 MPa with a factor of 2	19.812 MPa <= 55 MPa	Within known science	DWD-DEC-001, to confirm item 8
S20	Brine-wetted parts survive hot CaCl2 solution: tray insides, floors, absorbers and drip screens in black PTFE-coated aluminium, the floor mesh PTFE-coated glass fibre. Was bare aluminium and a stainless woven mesh, which pit in chloride brine and, above about 60 C, risk chloride stress corrosion cracking	Chemistry	Electrochemistry and batteries	Chloride pitting of aluminium; chloride stress corrosion cracking of 304 and 316 is rare below 60 C (SSINA), the beds run to about 110 C; PTFE is inert to CaCl2 and rated to 260 C	Bare metal fails the 300-cycle life; coated parts within known science, coating damage carried in the design FMEA	Within known science	DWD-DEC-001 D17
S21	The composite keeps 80 % or more of its capacity over 300 cycles	Chemistry	Materials and polymer chemistry	CaCl2 does not decompose below about 175 C; ordinary silica gel can crack when wetted with	Cycling test needed (TRL 4)	Plausible, to confirm at TRL 3	Open (TRL 4)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
				solution, so a water-resistant grade is specified (DWD-DEC-001, to confirm item 1)			
S22	The lumped bed model holds although heat enters at the plate on top: about 16 K drop through the 9.3 mm bed at noon; desorbing at the mid-depth temperature the yield is 0.41 L per day [C12]	Mathematics	Numerical method	Conduction through a packed bed, k about 0.2 W/(m K)	Yield falls from 0.44 to 0.41 L per day; PR-001 at risk under this sensitivity	Plausible, to confirm at TRL 3	DWD-CAL-001 C12
S23	Condensate chloride stays below 250 mg/L: brine cannot reach the plate (pores never fill, sumps, no line of sight)	Chemistry	Analytical chemistry	EU Directive 2020/2184 indicator value 250 mg/L; WHO taste threshold about 250 mg/L	Water test needed (TRL 4)	Plausible, to confirm at TRL 3	Open (TRL 4)
S24	Collected water is free of harmful microbes once treated; the device itself is not a treatment step	Biology	Microbiology	Condensate is near distilled but picks up dust and microbes from surfaces and the bottle; boiling or disinfection before drinking	Documents require treatment before drinking; laboratory test needed (TRL 4)	Plausible, to confirm at TRL 3	Open (TRL 4)
S26	Pessimistic combination of the calculation's uncertainties (bed temperature drop and half the light through the absorber holes absorbed): 0.37 L per day [C13] against 0.44 on the lumped model	Mathematics	Statistics and uncertainty	Independent review: the sensitivities were never combined	PR-001 not met on paper; range to be narrowed by measuring absorptance and isotherms (TRL 4)	Plausible, to confirm at TRL 3	DWD-CAL-001 C13
S27	Parts near the hot interior stay inside their ratings on a	Chemistry	Materials and polymer chemistry	Gap air at about the mean of bed and air (about 85	Supplier ratings to confirm	Plausible, to	Open (TRL 4)

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	peak day (bed about 133 C): plywood and PIR walls, EPDM seals, silicone, PTFE coatings			C); fan, battery and electronics are outside the box; PTFE 260 C, silicone and EPDM typically above 130 C, PIR and plywood ratings to confirm with suppliers	(DWD-DEC-001, to confirm item 12)	confirm at TRL 3	
S25	The yield simulation conserves energy: unaccounted heat at dusk 0.08 kWh is sensible heat left in the bed	Mathematics	Units and dimensional analysis	Sensible heat of the bed, trays and absorbers (about 7 kJ/K) over its rise at dusk	0.0805704 kWh <= 0.1438 kWh	Within known science	DWD-CAL-001 C7
B1	Option B: a high-emissivity panel under a clear night sky runs about 10 K below the air	Physics	Optics and radiation	Net radiation to a sky of emissivity 0.75 (Berdahl and Martin) against convection at 5 W/(m2 K)	claimed 10 K, estimate 9.9 K (ratio 1.01)	Within known science	DWD-OPT-001
B2	Option B makes no water at the design point: the panel (10.1 C) stays above the dew point (6.0 C at 40 % RH; -0.6 C at 25 %)	Physics	Thermodynamics	Dew forms only on a surface below the dew point; field data show no condensation below about 71 % RH (Atashi et al., 2021)	5.98811 C <= 10.1 C	Within known science	DWD-OPT-001
C1	Option C: with the cloth at 80 C and the glazing at 50 C the brine dries to about 52 wt %, and takes up about 0.7 kg overnight at 40 % RH by natural convection	Chemistry	Physical chemistry (equilibria)	Regeneration ends where aw x psat(cloth) = psat(glazing): 52 wt %; night equilibrium 40.1 wt %; mass transfer 2.1 mm/s on 3.2 m2 of cloth (estimate)	claimed 0.7 kg, estimate 0.722 kg (ratio 0.97)	Within known science	DWD-OPT-001
C2	Option C's yield hinges on the cloth temperature: with the cloth at 70 C the brine dries only	Physics	Thermodynamics	Same balance with psat(50 C) / psat(70 C): 44.0	claimed 0.27 kg, estimate 0.272 kg (ratio 0.99)	Within known science	DWD-OPT-001

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	to about 44 wt % and takes up about 0.27 kg overnight			wt % against 40.1 wt % at night			
C3	Option C collects about 0.2 L per day (cloth at 70 C) to 0.5 L per day (cloth at 80 C) at the design point, far below the 2.5 L per m2 per day reported for unstated test conditions (Kabeel, 2007)	Physics	Thermodynamics	Solar still collection efficiency about 0.7 (estimate): 0.19 to 0.51 L per day	claimed 0.5 L/day, estimate 0.505 L/day (ratio 0.99)	Within known science	DWD-OPT-001

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3**, **Fails, fix proposed**, **Fails**. Rows S5, S12 and S20 were **Fails, fix proposed** at the first check, and S9 and S18 after the independent review; all five are closed by the fixes below.

Fixes made under the standing instruction

Item	What failed	Fix (DWD-DEC-001)	Result
S5, absorber	The trays are open pans and 93 % of their footprint is the sorbent bed; the beads are white (absorptance about 0.3, an estimate), not the 0.95 black surface DWD-CAL-001 up to v0.7 assumed. Rerun, the design makes 0.01 L per day with the bed at 64 °C at noon (C11)	D16: a 1 mm black PTFE-coated perforated aluminium plate (3 mm holes at 6 mm pitch, 22.7 % open) resting on each bed; night air passes through the holes	Transmittance-absorptance 0.70 (C11)
S9, tray floor (independent review)	The floor was a 50 % perforated sheet under a fine mesh, passing about 25 % of the bed area to the vapour, against the 60 % DWD-CAL-001 took; corrected, the design makes 0.36 L per day	D19: a PTFE-coated glass-fibre mesh, about 1.8 mm aperture, on seven 6 x 1.5 mm bars on edge, 59 % open	0.44 L per day at 40 % RH, 0.31 L at 25 % (C3, C4)
S20, brine-wetted materials	Bare aluminium pits in chloride brine; the stainless woven floor mesh risks chloride stress corrosion cracking above about 60 °C (SSINA), and the beds run to about 105 °C	D17: black PTFE coating on the tray insides, floor bars, absorbers and drip screens; PTFE-coated glass-fibre floor mesh	Coated parts inert to $CaCl_2$ and rated to 260 °C; coating damage carried in DWD-RSK-001 (F2)
S12, condenser emissivity	DWD-CAL-001 (C8) gives the finned underside an emissivity of 0.85; the fins were bare aluminium (about 0.1)	D18: fins and plate underside painted with a matt, high-emissivity exterior paint (about 0.9); option B's best idea grafted onto option A	The C8 condenser figures hold; peak rise 11.9 K (C2)
S18, anchors (independent review)	Each anchor must hold 96 N at 20 m/s against a 100 N rating, and 234 N in a 25 m/s gust (H5)	D20: anchors rated 250 N or more each in the site soil; design wind speed a 25 m/s gust (decided by Amish, 2026-10-25)	Plausible until the rating in the site soil is confirmed

None of the fixes changes the working principle, the pitch or the scope. Together they add an estimated $56 (cost $695 against the $520 target). They do not restore PR-001, which is for Amish (DWD-DEC-001, open item 2). All are recorded in DWD-DEC-001 as decided under Amish's standing instruction of 2026-10-23, and he can reverse them.

Fields not relevant

None. Every field is touched: physics (psychrometrics, fluid flow through the bed, radiation and optics, heat transfer, diffusion, statics), chemistry (salt-solution equilibria, thermochemistry, corrosion, polymer ratings, analytical limits), biology (microbial quality of the water, burn thresholds), mathematics (energy closure and the lumped-bed approximation) and engineering (power budget, control rule).

Concept options

- Option A, solar desiccant box: rows S1 to S27. Within known science after the fixes; ten rows plausible; collects 0.44 L per day, below PR-001.

- Option B, radiative dew condenser: within known science (B1, B2). The panel runs about 10 K below the air, but at 20 °C and 40 % RH the dew point is 6.0 °C, so it makes no water at the design point. It is not dropped on STEM grounds; its yield is scored in DWD-OPT-001.
- Option C, liquid-desiccant glazed still: within known science (C1 to C3), with a yield of 0.2 to 0.5 L per day that depends strongly on how hot the cloth gets. The published 2.5 L per m² per day (Kabeel, 2007) does not carry over to the design point.

Independent review

A second agent (a separate Claude session run on 2026-10-25 in a folder holding only this STEM check, the calculation note DWD-CAL-001, the product requirements DWD-PRS-001 and `sizing.py`, with none of the author's reasoning) was asked to refute the headline results and every "Within known science" verdict. It rebuilt the day model with stubbed geometry and reproduced the script for the version it saw (0.50 against 0.51 L per day). What it found and what changed:

Row or item	Finding	Verdict	Change
S9, vapour transport	Stefan arithmetic upheld, but the 60 % open share of the tray floor was a stale input: a 50 % perforated sheet under a fine mesh passes about 25 to 30 %	Refuted (input)	D19 (open floor of mesh on bars, 59 %); yield 0.36 L per day as it was, 0.44 L with the fix
PR-001 headline	2 % margin on unmeasured estimates; sensitivities never combined; any one plausible correction drops it below target	Weakened	S26 and C13 added (0.37 L per day); PR-001 now stated as not met; recovery to Amish
S5, absorber	Arithmetic upheld; 0.70 for light through the holes unmeasured, perhaps lower; no glazing angle or soiling loss	Weakened	Estimates listed in DWD-CAL-001 limits; C13 uses 0.5
S13, stagnation	Run at 785 W/m², and the bed compared with the glazing rating; at 1000 W/m² the bed reaches 135 to 149 °C; hot materials other than the glazing unexamined	Weakened	D6 added (133 °C bed, 106 °C inner skin); S13 now checks the skin; S27 added (plausible)
S18, wind	6 % margin at 20 m/s; anchor load about 200 N at 25 m/s	Weakened	D20 (250 N anchors); design wind speed decided by Amish on 2026-10-25: a 25 m/s gust until the site value is known
S11, condenser	Isolated-plate correlation optimistic for fins under a hot tilted floor; PR-005 fails at 30 % less convection and half the radiation	Weakened	S11 plausible; PR-005 at risk
S15, S16, salt	One-night figure plausible; protection kinetic; cut-out 1 point below pore filling, inside sensor tolerance	Weakened	Lower cut-out proposed to Amish (DWD-DEC-001, item 3); decided 2026-10-25: 65 % RH, restart below 60 %
S8, efficiency	High for a single stage	Weakened	S8 plausible
S22, bed gradient	Optimistic end of the gradient effect	Weakened	Kept plausible, closed by C12 with C13 alongside
S1, S2, S3, S4, S6, S7, S10, S14, S20, S25	Upheld (S10's value and limit labels clarified; S6 and S25 close by construction, not independently)	Upheld	S10 wording

The reviewer did not check S19, S21, S23, S24 or options B and C beyond reading them.

Sources

- Alduchov, O. A., and R. E. Eskridge. "Improved Magnus Form Approximation of Saturation Vapor Pressure." *Journal of Applied Meteorology* 35 (1996): 601 to 609 (S1, psychrometrics in `stem_check.py`).
- Conde, M. R. "Properties of Aqueous Solutions of Lithium and Calcium Chlorides." *International Journal of Thermal Sciences* 43 (2004) (S2, S15, C1 to C3).
- Ergun packed-bed equation and laminar vertical-plate natural convection, Nu = 0.59 Ra^0.25: standard heat- and mass-transfer texts (S4, S11).

- Kim, H., et al. "Adsorption-Based Atmospheric Water Harvesting Device for Arid Climates." *Nature Communications* 9 (2018): 1191. https://doi.org/10.1038/s41467-018-03162-7 (checked 2026-10-25): the sorbent layer is backed by a black absorber of solar absorptance about 0.95 (S5).
- LaPotin, A., et al. "Dual-Stage Atmospheric Water Harvesting Device for Scalable Solar-Driven Water Production." *Joule*, 2021 (S8).
- Specialty Steel Industry of North America. "Chloride Stress Corrosion Cracking." https://www.ssina.com/education/corrosion/chloride-stress-corrosion-cracking/ (checked 2026-10-25): chloride stress corrosion cracking of 304 and 316 is rare below 60 °C (S20).
- World Health Organization, chloride background document, and Directive (EU) 2020/2184, Annex I (S23); see DWD-REG-001.
- Berdahl, P., and M. Martin. Clear-sky emissivity as a function of dew point (sky emissivity in B1).
- Atashi, N., et al. *Hydrology and Earth System Sciences* 25 (2021): 4719 to 4740. https://hess.copernicus.org/articles/25/4719/2021/hess-25-4719-2021.html (checked 2026-10-25) (B2).
- Kabeel, A. E. *Renewable Energy* 32, no. 1 (2007): 157 to 172. https://ideas.repec.org/a/eee/renene/v32y2007i1p157-172.html (checked 2026-10-25) (C3).
- DewDrive calculation note DWD-CAL-001 v0.8 for every tagged value.